Lecture 2: Pattern matching and ELIZA

PSYC 51.07: Models of language and communication

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Recap from Lecture 1

Key ideas

- **Consciousness is complex:** multiple types, hard to define
- **Language $\neq$ Thought:** but they interact in interesting ways
- **Grounding matters:** meaning comes from experience

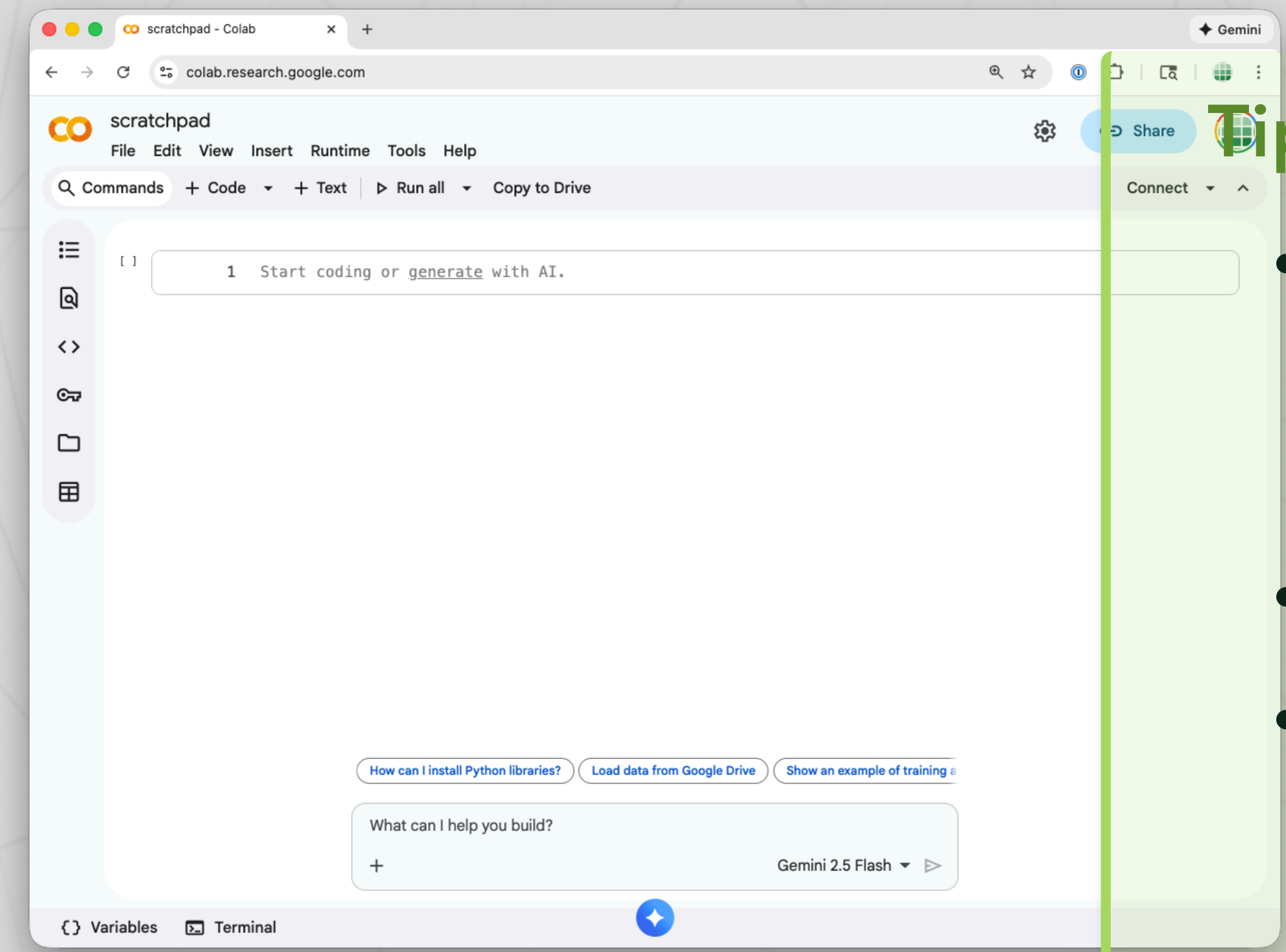
However...

Does the "experience" need to be first-hand? Can we "share" experience with another person? Or with a machine? What might that look like?

Today's focus

Pattern matching can create powerful *illusions* of understanding, even without any "real" comprehension.

Side note: follow along with Google Colab!



Tip

- Go to colab.research.google.com
- Click "New notebook"
- Click to start new `text` or `code` cells
- Copy and paste code

Creating the illusion of experience and understanding

Humans

- Derive meaning from experience
- Connect words to memories, emotions, senses
- Understand context and nuance

Computers

- Manipulate strings (sequences of characters)
- Have no direct experience of the world
- Process symbols without inherent meaning

String manipulation basics

Text processing is the foundation of computational linguistics:

- **Finding:** Locate patterns within text
- **Replacing:** Substitute one pattern for another
- **Extracting:** Pull out specific parts of text
- **Transforming:** Convert text to different formats

```
text = "Hello, how are you today?"

```

Why string manipulation matters

User input → Pattern matching → Response generation → Output

Basic conversational AI pipeline

> **Note**
>
> Every conversational AI system, from ELIZA to ChatGPT, fundamentally processes text through some form of pattern matching, though with vastly different sophistication.

Regular expressions

Regular expression (regex)

A sequence of characters that defines a search pattern. Regular expressions provide flexible, powerful pattern matching for text processing.

Key syntax

Symbol	Meaning	Example
`.`	Any character	`h.t` matches "hat", "hit", "hot"
`*`	Zero or more	`ab*c` matches "ac", "abc", "abbc"
`()`	Capture group	`(hello)` captures "hello"
`|`	Alternation	`cat|dog` matches "cat" or "dog"

Regular expressions in Python

```python
import re

text = "I am feeling very happy today"

# Simple pattern matching
if re.search(r"happy|sad|angry", text):
    print("Found an emotion!")

# Capture groups - extract parts of a match
match = re.search(r"I am feeling (.*?) today", text)
if match:
    emotion = match.group(1)  # "very happy"

# Substitution
new_text = re.sub(r"I am", "You are", text)
# "You are feeling very happy today"
```

Meet ELIZA

Historical context

ELIZA was created by Joseph Weizenbaum at MIT in 1966. It was one of the first programs to attempt natural language processing.

Why a Rogerian therapist?

- Non-directive therapy style
- Reflects statements back to patient
- Asks open-ended questions
- Avoids making claims

Key insight

- Rogerian style requires no real knowledge
- Simply reflect and rephrase
- Let the human do the "heavy lifting"

Reading: Weizenbaum (1966)

Required reading

Weizenbaum, J. (1966). ELIZA—A computer program for the study of natural language communication between man and machine. *Communications of the ACM*, 9(1), 36-45.

Pay attention to:

- How does ELIZA select responses?
- What are "scripts" in ELIZA's architecture?
- Why did Weizenbaum choose the DOCTOR script?
- What did Weizenbaum observe about user reactions?

Live demo

Try it yourself!

ELIZA Demo

Discussion prompts:

- What does ELIZA do surprisingly well?
- What reveals its limitations?
- Can you "trick" ELIZA? How?
- What kinds of inputs break the illusion?

Warning

Try to have a "real" conversation. At what point does the illusion break down?

The ELIZA effect

The ELIZA effect

The tendency to unconsciously assume that computer behaviors are analogous to human behaviors; to attribute human-like understanding to programs that merely simulate it.

Weizenbaum's observation

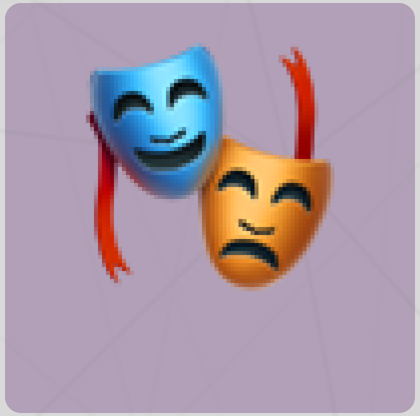
Weizenbaum was surprised (and disturbed) by how quickly users became emotionally involved with ELIZA. His secretary reportedly asked him to leave the room so she could have a private conversation with the program!

Why do we anthropomorphize machines?


Language cues


Pattern recognition


Social instincts


Theory of mind

> **Note**
>
> Humans are social creatures. We evolved to detect minds and intentions, and we often over-apply this tendency, even to clearly non-conscious entities.

Discussion

Reflect on your experience

Have you experienced the ELIZA effect with modern AI systems (ChatGPT, Claude, Siri, Alexa)?

Consider:

- When have you felt like an AI "understood" you?
- What broke the illusion?
- What is the difference between *seeming* intelligent and *being* intelligent?

The hard question

How would we know if an AI truly understood us?

Up next

Lecture 3 (Thursday X-hour)

How ELIZA actually works

- Complete architecture walkthrough
- Pattern matching and response selection
- The role of scripts and keywords
- We will build it ourselves!

Prepare for next time

- Finish reading Weizenbaum (1966)
- Play with the ELIZA demo
- Think about: What would YOU add to ELIZA?

Key takeaways

1. **String manipulation is foundational:** All text-based AI builds on finding, replacing, and extracting patterns
2. **Regular expressions are powerful:** Flexible pattern matching enables sophisticated text processing
3. **ELIZA demonstrated the power of simplicity:** A few clever rules can create convincing illusions
4. **The ELIZA effect is real:** We naturally anthropomorphize systems that use language
5. **Seeming $\neq$ Being:** Appearing intelligent does not require actual understanding

Questions? Want to chat more?

Email me

Join our Discord

Come to office hours

Tip

This course will move *very* quickly. **Please** reach out if you have questions, comments, concerns, or just want to chat!